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Data-driven analysis in 3D concrete printing: predicting and optimizing construction mixtures

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Abstract

Accurately predicting 3D concrete printing (3DCP) properties through the utilization of machine learning holds promise for advancing cost-effective, eco-friendly construction practices that prioritize safety, reliability, and environmental sustainability. In this study, a comprehensive exploration of seven regression models was undertaken, complemented by the application of Bayesian optimization techniques to forecast critical metrics such as compressive strength, pump speed, and carbon footprint within the realm of 3DCP technology. Drawing upon a compilation of various 3DCP mixtures sourced from existing literature, an intricate carbon footprint calculation methodology was devised, resulting in the establishment of a bespoke database tailored to the study's objectives. The performance evaluation of the developed models was conducted through the analysis of key statistical indicators, including R^2 , RMSE, MAE, and Pearson correlation. To enhance the robustness and generalizability of the models, a rigorous tenfold cross-validation strategy coupled with a strategic introduction of noise was employed during the validation process. The incorporation of Shapley Additive Explanations (SHAP) analysis provided insightful interpretability into the predictive capabilities of the models, enabling a nuanced understanding of the underlying relationships between input variables and target outputs. Furthermore, the application of multi-objective optimization techniques facilitated judicious decision-making processes, enabling the identification of optimal 3DCP mixture compositions that concurrently enhance performance metrics, reduce operational costs, and mitigate CO₂ emissions.

Keywords Machine learning, 3D concrete printing, Compressive strength, Pump-speed, Carbon footprint, Bayesian optimization, Shapley additive explanations (SHAP), Multi-objective optimization

1 Introduction

In recent years, the realm of manufacturing has undergone a considerable transformation due to the advent of 3D printing technology that has revolutionized the construction industry by facilitating the precise and efficient fabrication of multistory buildings, horizontal structural components, and a wide array of construction elements. Consequently, architects and builders now have an expanded repertoire of possibilities for crafting

distinctive and functional architectural structures (Jeong et al., 2019). There are many impressive examples of what 3D printing can achieve in architecture and construction, including Andrey Rudenko's 3D-printed castle in the United States (Rudenko, 2014), a two-story building constructed by Apis Cor in the UAE in 2019 (Yusuf, 2017), a 3D-printed retractable bridge in China (Everett, 2021), a 3D-printed house in Nantes by the University of Nantes and the Nantes city council in France (Cowan, 2018), and a community village in Austin created by ICON (Nielsen, 2020). These exemplary instances underscore the transformative potential of 3D printing technology in transcending the constraints of traditional construction methodologies.

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However, the field of 3DCP technology also encounters significant challenges, particularly regarding the optimization of material properties such as strength and workability, while simultaneously minimizing waste and environmental impact. A key hindrance is the inadequacy of comprehensive datasets, impeding the development of precise predictive models. Recent research has illustrated that machine learning models can significantly enhance the accuracy of predicting material properties in 3DCP; however, the issue of data scarcity persists (Rehman et al., 2024; Wang et al., 2024). Moreover, the real-time monitoring and adjustment of 3D printing parameters, including print speed and material flow rate, are imperative for enhancing print quality and consistency (Ali et al., 2023). The integration of machine learning models into the 3DCP process, particularly those employing techniques such as support vector machines and Gaussian Process Regression, has presented encouraging outcomes in optimizing tensile and flexural strength in 3DCP applications (Ali et al., 2023).

The characteristics of 3DCP are the subject of ongoing investigation within the academic community. Researchers are actively exploring various aspects of 3DCP technology, with a particular focus on the compressive strength of 3DCP structures. Day (1994) and Matulić (2016), for instance, have delved into the relationship between the size of solid samples and their capacity to withstand compression. Their findings suggest that larger specimens tend to exhibit lower compressive strength. Del Viso et al. (2008) conducted a study on the dimensions and shapes of cubic and cylindrical solid samples, which were made of high-strength concrete, to assess their performance in compression tests. Their experimental results indicated that cylindrical samples consistently demonstrated a uniform average ultimate tensile strength (UTS). In a similar vein, Aramburu et al. (2022) investigated the mechanical behavior of annular 3D-printed cylindrical specimens and how it impacts the compressive strength of concrete structures. Beyond the physical properties of 3DCP, researchers have also focused on optimizing the material composition for enhanced performance. For example, Xu et al. (2022) examined the influence of FA (fly ash) and GGBFS (ground granulated blast furnace slag) on the compressive strength of cement-based 3D printing materials. Meanwhile, Chen et al. (2020) proposed the addition of diatomite and retarders to improve the printability of 3D-printed sulphoaluminate cement paste. In the realm of practical application, the flow characteristics of 3DCP during pumping and its rapid stiffening upon hardening are crucial considerations for efficient construction. Pump speed plays a significant role in this process. Lee et al. (2021) observed that the volume of each layer of

3DCP materials varies depending on the nozzle speed and printing duration. Similarly, Tao et al. (2022) analyzed the interplay between pumping and deposition processes in 3D concrete printing. Furthermore, researchers such as Mohan et al. (2021) and Moeini et al. (2022) have emphasized the importance of pump speed in conjunction with other factors, conducting tests on various velocities and rheological parameters to gain insights into the pumping behavior of printable concrete.

The integration of machine learning (ML) techniques within the realm of 3DCP holds substantial promise for advancing the field. Machine learning has the capability to meticulously analyze vast datasets, recognize underlying patterns, and enhance the precision and efficiency of the printing process. Moreover, it can facilitate advancements in material selection and design, ultimately enabling the fabrication of intricate structures with superior characteristics. The applications of 3D concrete printing technology span various domains, encompassing the refinement of sustainable construction topologies, molecular-level material simulation, and the optimization of tool paths during the early hydration phase of concrete (Wangler et al., 2019). Additionally, previous research has shown that incorporating waste materials like marble and granite powder into concrete can reduce construction costs and improve mechanical properties (Ullah et al., 2021). This interdisciplinary field is relatively nascent and continuously evolving (Wangler et al., 2019), with significant research progress seen in ML-driven extrusion-based 3D printing of concrete (Li et al., 2019; Ribeiro et al., 2021) and ML-guided studies on concrete materials (Das et al., 2019; Lapeyre et al., 2021; Van Dao et al., 2019). The amalgamation of machine learning and 3D concrete printing presents a unique opportunity to deepen comprehension and hasten progress in the field of 3DCP. For example, Lao et al. (2020) employed ML to simulate printed concrete mix structures via extrusion to improve surface smoothness. The amalgamation of machine learning and 3D concrete printing presents a unique opportunity to deepen comprehension and hasten progress in the field of 3DCP (Bellini, 2002; Comminal et al., 2018; De Larrard & Roussel, 2011; Spangenberg et al., 2012; Vasilic et al., 2016). Furthermore, Marczyk et al. (2021) delved into the potential of artificial intelligence and machine learning methodologies in formulating efficient concrete extrusion materials for 3D printing, particularly focusing on blends incorporating metakaolin, fly ash, and cement. Such data-driven approaches have the capacity to streamline both the cost and time requirements associated with research and development endeavors. Additionally, Izadgoshasb

et al. (2021) devised frameworks for predicting the compressive strength of 3D-printed cement utilizing a multi-objective optimization algorithm that combines artificial neural networks and multi-objective grasshopper optimization algorithms. Lastly, Bagheri and Cremona (2020) built four supervised ML decision tree algorithms to categorize 114 instances of boron-based geopolymer 3DCP concrete into four groups based on compressive strength variations, and subsequently conducted statistical evaluations on each model.

This study delves into the application of machine learning to bolster the capabilities of 3D concrete printing in advancing environmentally conscious construction practices. The central objective is to evaluate the efficacy of diverse model architectures in accurately forecasting parameters such as compressive strength, pump speed, and carbon footprint, leveraging Bayesian optimization and fine-tuning strategies to refine model performance. Drawing on a distinctive dataset sourced from 32 references, the study constructs robust predictive models to characterize the properties of 3D concrete printing. Furthermore, through SHAP analysis, the study illuminates the impact of different input variables on model predictions, providing valuable interpretability. By integrating a multi-objective evolutionary algorithm (MOEA), the research streamlines optimization efforts encompassing performance, cost efficiency, and ecological sustainability.

2 Methodology

The present study employs a methodological framework as illustrated in Fig. 1, which delineates a systematic approach to analyze the predictive performance of seven ML models. These models consist of four boosting techniques [namely Gradient Boosting Regressor (GBR), XGBoost, LightGBM, and AdaBoost], two tree-based models (Random Forest and Decision Tree), and a multi-layer perceptron. The experimental dataset utilized in this research pertains to 3DCP mixtures sourced from existing literature, with a focus on predicting variables such as compressive strength, pump speed, and carbon footprint. To address the prediction of carbon footprint, an initial methodology is established to collate the requisite data for the dataset. Once a comprehensive dataset encompassing compressive strength, pump speed, and carbon footprint is compiled, the analytical process commences with an initial data split. Subsequently, four boosting models are developed through Bayesian optimization to fine-tune model-specific parameters. To prevent overfitting, additional strategies, including fine-tuning, k-fold cross-validation, and bootstrap aggregating techniques, are integrated into the modeling process. Moreover, specific regression metrics are employed to evaluate the performance of the models. The analysis extends to a detailed evaluation of the best-performing model, conducted through quantitative analysis. The SHAP method is utilized to dissect the impact of various factors on the model predictions. Furthermore, a

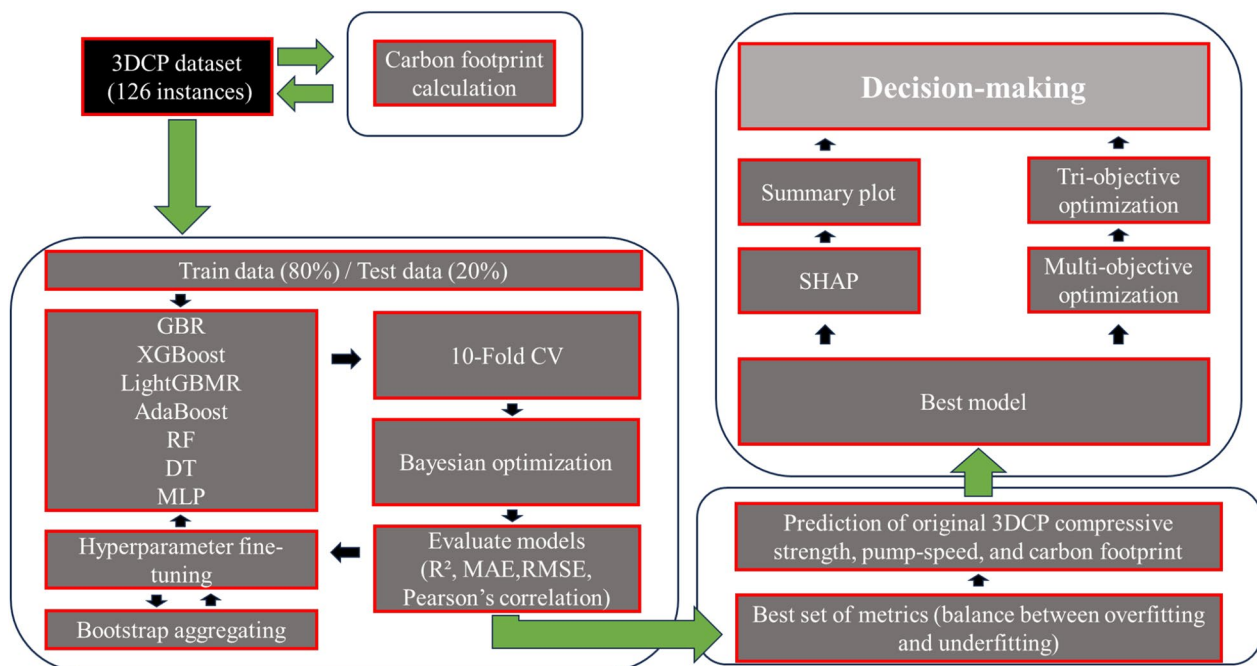


Fig. 1 Flowchart of the proposed analysis

multi-objective evolutionary algorithm (MOEA) is leveraged to optimize and balance the multiple objectives under consideration. This inclusive approach facilitates the achievement of efficient and informed decision-making within the analytical framework.

2.1 Data sample analysis

The present forecasting model is built upon an analysis of a dataset consisting of 126 data points sourced from a variety of peer-reviewed scientific publications (Baz et al., 2020; Bos et al., 2020; Chen et al., 2019, 2021; Kazemian et al., 2017; Khalil et al., 2017; Kruger et al., 2021; Le et al., 2012; Lee et al., 2019; Long et al., 2019; Marchment et al., 2019; Mazhoud et al., 2019; Mechtcherinea et al., 2019; Murcia et al., 2020; Nerella & Mechtcherine, 2016; Nerella et al., 2019; Paul et al., 2018; Pham et al., 2020; Rahul et al., 2019a, 2019b, 2022; Rehman et al., 2020; Shakor et al., 2017; Ting et al., 2019; Van Der Putten et al., 2019; Wang et al., 2020; Weng et al., 2018; Yu et al., 2021; Zhu et al., 2019). The complete database is detailed in Table S1, with corresponding references listed in Table S2, both of which can be found in the supplementary section. This dataset encompasses ten distinct input variables: cement (C, kg/m³), water (W, kg/m³), fine aggregate (FineAgg, kg/m³), fly ash (FA, kg/m³), silica fume (SF, kg/m³), superplasticizer (SP, kg/m³), slag (S, kg/m³), admixture (Admix, kg/m³) that in this case includes viscosity modifying agent (VMA), high range water reducer agent (HRWRA), and regular retarders, and fibers (F, kg/m³). The output variables considered in the Machine Learning model are Compressive strength (CS, MPa), pump-speed (PS, mm/s), and Carbon footprint (CF, tCO₂/m³). In Table 1 the statistical results for both input and output variables in the dataset are provided. Notably, pump-speed exhibited a small percentage of missing values (less than 10%), which were addressed through the implementation of the missForest method. MissForest, an iterative imputation technique leveraging Breiman's random forests (Stekhoven, 2012), offers advantageous features including managing complex interactions and non-linearity through a non-parametric approach, accommodating various types of missing data, and adapting well to datasets with high dimensions. The selection of missForest was influenced by its proven efficacy demonstrated in several studies, where it showed superiority over other imputation methods, particularly in datasets with complex interactions and nonlinear relationships (Okafor & Delaney, 2021; Stekhoven, 2012; Waljee et al., 2013).

2.1.1 Carbon footprint calculation

This study employed a comprehensive methodology to assess the carbon footprint (CF) of the materials used

Table 1 3D concrete printing attributes

Variables	Min	Max	Mean	Stdev
Cement (kg/m ³)	289.00	1500.00	581.15	214.48
Water (kg/m ³)	123.50	550.00	246.64	76.80
Fine Agg (kg/m ³)	0.00	1612.00	970.96	313.02
Fly Ash (kg/m ³)	0.00	1000.00	115.70	205.34
Silica Fume (kg/m ³)	0.00	300.00	42.60	69.14
SP (kg/m ³)	0.00	18.80	6.71	5.68
Slag (kg/m ³)	0.00	376.30	21.68	83.49
Admixture (kg/m ³)	0.00	25.00	2.76	4.37
Fibers (kg/m ³)	0.00	78.50	2.30	10.17
Pump-Speed (mm/s)	0.08	200.00	53.20	50.22
Compressive Strength (MPa)	13.00	110.00	62.34	23.78
Carbon Footprint (tCO ₂ /m ³)	878.71	2945.02	1679.88	572.88

in 3D printed concrete, an essential step in appraising the environmental ramifications of diverse construction materials. The carbon footprint signifies the aggregate greenhouse gas emissions, predominantly encompassing carbon dioxide (CO₂), measured in kilograms per cubic meter of the material.

The CF for each material was computed using the Eq. (1).

$$CF = CEF \times \text{Density} \quad (1)$$

where, CF (kgCO₂/m³) represents the carbon footprint of the material, CEF (kgCO₂/kg) refers to the carbon emission factor, which quantifies the CO₂ emissions per kilogram of material, and density (kg/m³) is the mass per unit volume of the material.

The carbon emission factors (CEF) and material densities used in this study were obtained through a comprehensive review of published literature, established Life Cycle Assessment (LCA) databases, and material property references. Primary data sources, such as the ecoinvent database, were consulted to ensure the accuracy and reliability of the data utilized for carbon footprint assessments. Noteworthy references, including Jones (2019) for the CEF of ordinary Portland cement and Helsel et al. (2016) for its density, were instrumental in deriving key parameters. For example, the CEF for ordinary Portland cement was determined to be 0.9 kgCO₂/kg (Jones, 2019), while its density was recorded as 3150 kg/m³ (Helsel et al., 2016). Similarly, alternative materials, such as fly ash, silica fume, and superplasticizers, were assigned appropriate CEF values based on information derived from the ecoinvent database and additional supporting literature sources.

Utilizing Eq. (1), the CF was calculated for each material by the multiplication of its corresponding CEF by its

respective density. The emission factors, densities, and resultant CF values for the various materials integrated within the composition of the 3D printed concrete amalgam are succinctly outlined in Table 2. For instance, the CEF for water amounted to 0.00046 kgCO₂/kg, coupled with a density of 1000 kg/m³, yielding a CF of 0.46 kgCO₂/m³. Similarly, fly ash exhibited a CEF of 0.027 kgCO₂/kg and a density of 810 kg/m³, producing a CF of 21.87 kgCO₂/m³. The detailed calculations for all materials, including fine aggregates, fibers, and admixtures, are presented in Table 2.

2.1.2 Data characterization and correlation analysis

The dataset presents the distribution range of each feature, illustrated in Fig. 2 through joint plot kernel density estimation (KDE), providing a visual depiction of the data in the form of histograms. Furthermore, a color map correlation matrix, featured in Fig. 3, has been devised to illuminate the interrelations among different pairs of input variables for regression modeling, utilizing correlation coefficients. A pronounced positive correlation coefficient present in a specific cell signifies a strong association between the respective input parameters. The correlation matrix reveals varying strengths of correlations among different parameters. Notably, there exists a substantial positive correlation between cement and water, whereas admixture and silica fume exhibit a notably weak correlation.

2.2 Machine learning models background and framework

The overview of the employed ML models in this study can be found in the supplementary information.

2.2.1 Bayesian optimization

In order to optimize the hyperparameters of the MLP model, we utilized the Bayesian Optimization technique from the GPyOpt library. Bayesian optimization, a probabilistic approach employed for function minimization, is commonly utilized for hyperparameter tuning

in machine learning tasks (Sameen et al., 2019). This method comprises various components, including evaluating the loss on the validation set with a specific set of hyperparameters, defining the search space encompassing the range of hyperparameter values to explore, utilizing an optimization algorithm to formulate the objective function and select hyperparameter values for evaluation, and assessing the objective function along with hyperparameter values and losses on the validation set to derive optimal outcomes (Rong et al., 2020). Moreover, Bayesian optimization employs a Gaussian process to compute an acquisition function for evaluating the objective function while maintaining a record of past iterations to guide the quest for optimal solutions (Awal et al., 2021).

The optimization process necessitated the delineation of an optimization domain with the ranges and types of hyperparameters to be optimized. In this instance, the hyperparameters comprised of the hidden layer size and dropout rate. Subsequently, the Bayesian optimization algorithm iteratively explored the parameter space, employing the acquisition function to strike a balance between exploration and exploitation. Ultimately, the refined hyperparameters yielded a configuration that optimized the R², resulting in enhanced performance and accuracy.

2.2.2 SHapley additive explanations (SHAP)

The Shapley value method, rooted in cooperative game theory principles, presents a flexible approach for ranking the significance of individual attributes in classification examples, thus providing valuable insights to aid decision-making (Shapley, 1952). Widely utilized in the field of machine learning, particularly in the evaluation of model predictions, scholars Lundberg and Lee (2017) underscore the potential of SHAP by drawing an analogy between input variables and "players" contributing to a "payout" in a game-like scenario. SHAP facilitates a quantitative evaluation of the distinct contribution of each "player" to the overall predictions, thereby unveiling

Table 2 Carbon emission factors and density

Ingredients of the Mix	CEF (kgCO ₂ /kg)	Density (kg/m ³)	CF (KgCO ₂ /m ³) CF = CEF * Density
Water	0.00046	1000	0.46
Fine aggregate	0.00438	2600	11.388
Fly ash	0.027	810	21.87
Silica fume	0.00031	2200	0.682
Superplasticizer	0.25	1200	300
Granulated blast furnace Slag	0.0265	1100	29.15
Admixture	1.67	1316	2197.72
Fiber (Glass fiber)	1.23	1900	2337

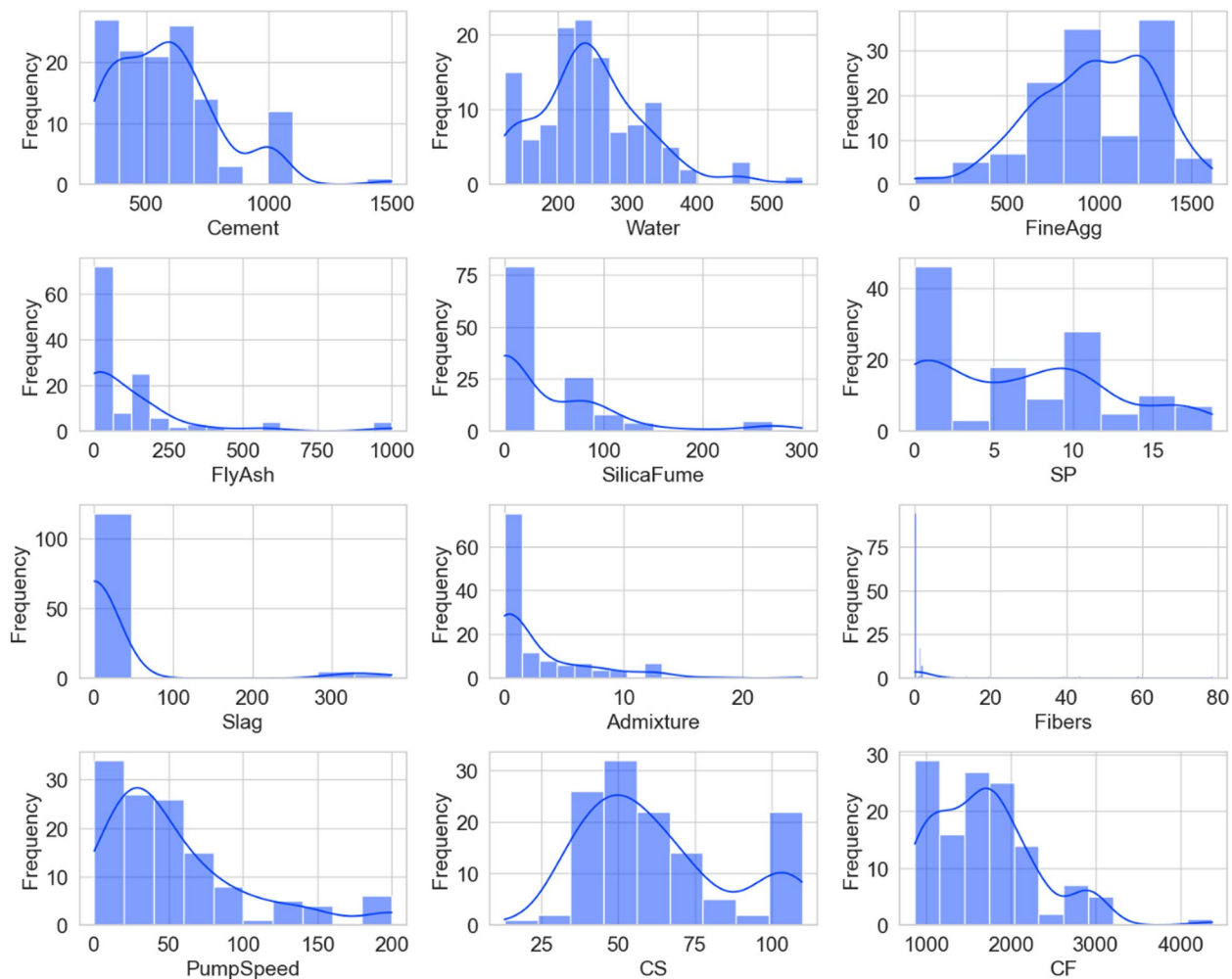


Fig. 2 Distribution plot of features

the underlying mechanisms of machine learning models on both a macroscopic and microcosmic scale (Sharma et al., 2020).

The significance of SHAP in deciphering machine learning model outputs cannot be overstated, as it ensures coherence in feature attributions. By computing Shapley values for each feature, it elucidates their impact on model predictions when incorporated, whether positively or negatively (Rathi, 2019). Furthermore, SHAP can be easily accessed through a Python toolkit specifically crafted for transparent machine learning, thereby rendering it practical for grasping intricate models and datasets (Mangalathu et al., 2020). A variety of methodologies have surfaced for leveraging the Shapley value in ascertaining feature importance. Within this framework, the Shapley value serves to evaluate the relevance of features by treating them as "players" in a game and assigning a metric of influence or importance to each (Kumar et al., 2020). The Shapley value ϕ_{ij} quantifies the contribution of

a specific feature j to the prediction for a given instance i , in relation to the average prediction of the dataset (Molnar, 2022).

In the year 2017, SHAP introduced an all-encompassing framework designed to elucidate predictions, thereby establishing itself as the exclusive method capable of delivering coherent and locally precise feature attribution grounded in probabilistic expectations. These SHAP values have emerged as a cohesive metric for assessing the significance of features by assigning unique importance values to individual features, thereby illustrating their respective contributions to model predictions when duly taken into account (Rong et al., 2020).

2.2.3 Multi-objective evolutionary algorithm (MOEA)

In contemporary discourse, the integration of MOEAs with various optimization techniques has garnered considerable attention for their capacity to enhance performance and tackle specific challenges within

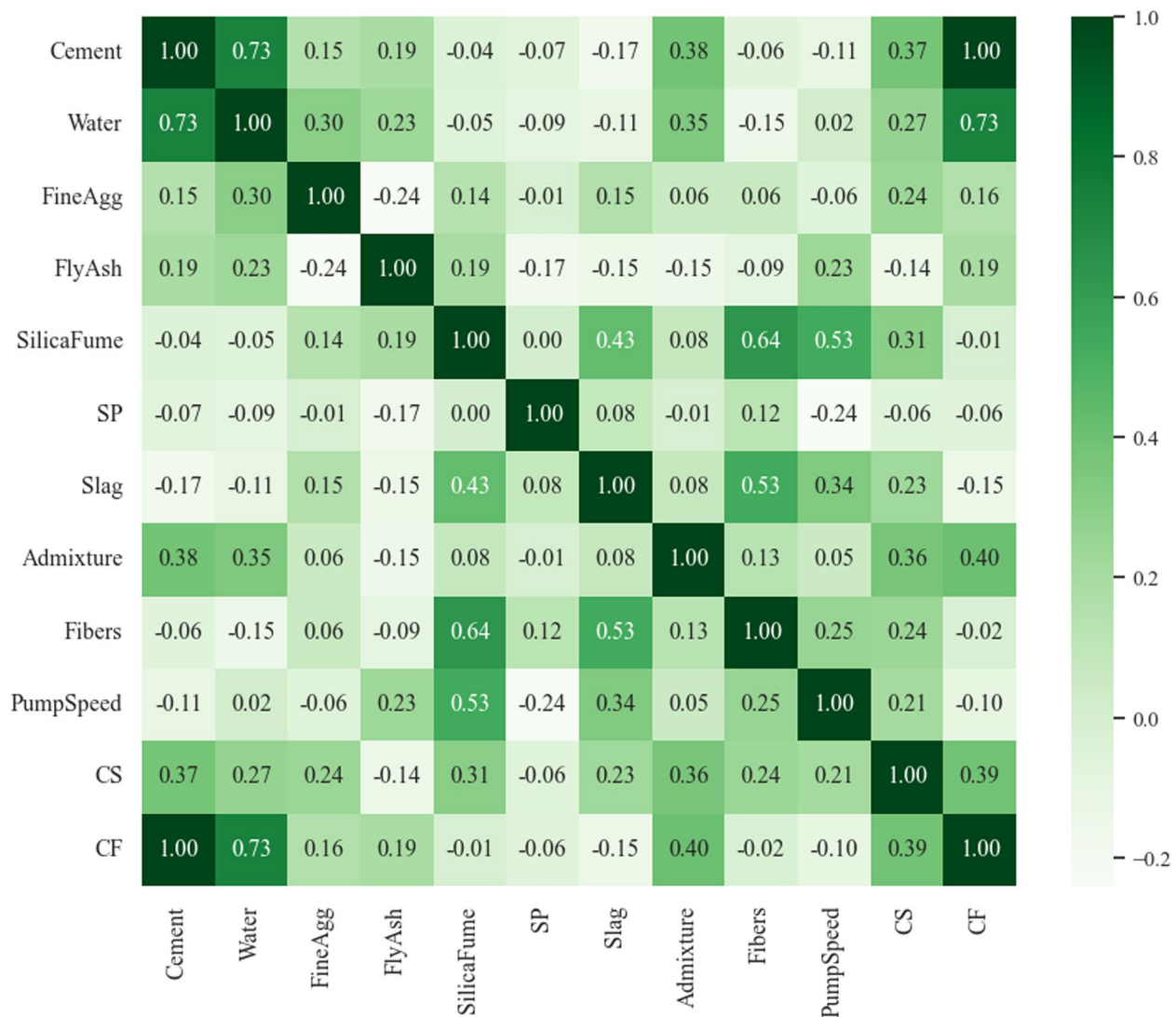


Fig. 3 Correlation matrix of the features

research and practical applications (Talbi, 2002). The evolution of MOEAs has permeated diverse domains, spanning from engineering design and chemistry to physics, medicine, and computer science (Coello Coello, 2006).

The utility of MOEAs lies in their ability to address optimization problems characterized by multiple conflicting objectives (Deb, 2001). These algorithms leverage evolutionary mechanisms, such as selection, crossover, and mutation, to iteratively evolve a population of solutions over successive generations (Coello Coello, 2006). Particularly within real-world contexts necessitating meticulous considerations of trade-offs among conflicting objectives, the relevance of MOEAs becomes pronounced (Zheng & Doerr, 2023). A salient attribute of

MOEAs is their proficiency in identifying a diverse set of non-dominated solutions, which form the Pareto-optimal front (Deb et al., 2002). This unique trait empowers decision-makers to explore a spectrum of optimal trade-offs rather than being constrained to a singular solution (Konak et al., 2006). Augmenting this, tools like parallel coordinates and interactive visual analytics have been developed to facilitate the examination of Pareto-optimal solutions (Deb et al., 2005). The versatility and robustness of MOEAs underscore their efficacy in addressing intricate optimization challenges situated within multi-objective environments (Deb et al., 2002).

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) has garnered widespread recognition and thorough examination in the realm of multi-objective

evolutionary algorithms due to its efficiency and efficacy in managing multiple conflicting objectives (Deb et al., 2002). NSGA-II utilizes a swift non-dominated sorting approach to categorize the population into various Pareto fronts, thereby facilitating the preservation of a spectrum of solutions (Deb, 2001). Additionally, it integrates a crowd distance mechanism to uphold solution diversity and prevent premature convergence (Srinivas & Deb, 1994). The algorithm's elitist strategy ensures the retention of the elite solutions for subsequent generations, thereby augmenting the convergence towards the authentic Pareto-optimal front (Deb et al., 2002). Its modest computational complexity renders NSGA-II particularly well-suited for handling large-scale optimization problems, thereby further elevating its prominence (Konak et al., 2006). The continual refinements and adaptations have solidified NSGA-II's status as a benchmark algorithm within the domain of multi-objective optimization (Deb & Goel, 2001).

2.2.4 Validation methods

This study employs four regression metrics to evaluate model performance. The first metric, the correlation coefficient (R^2), serves to quantify the degree of alignment between the data and the regression line, or the extent to which a linear model accounts for the variance in the response variable. The second metric, the mean absolute error (MAE), gauges the average magnitude of errors within a prediction set, without considering their direction, by evaluating the residual error at each sample point. The third metric, the root mean squared error (RMSE), is calculated by obtaining the square root of the average of the squared discrepancies between the predicted and actual values. Furthermore, Pearson's correlation coefficient (R) serves as a statistical measure to evaluate the linear association between predicted and observed values, thereby indicating how closely the predicted values correspond to the actual observations. Equations (2), (3), (4), and (5) provide the mathematical expressions for these four parameters, respectively.

$$R^2 = \frac{\left[\sum_{i=1}^n (y_i - \bar{y})(y'_i - \bar{y}') \right]^2}{\sum_{i=1}^n (y_i - \bar{y})^2 \cdot \sum_{i=1}^n (y'_i - \bar{y}')^2} \quad (2)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y'_i - y_i| \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (4)$$

$$R = \frac{\sum_{i=1}^n (y_i - \bar{y})(y'_i - \bar{y}')}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \cdot \sum_{i=1}^n (y'_i - \bar{y}')^2}} \quad (5)$$

In the equations, y_i refers to the original value for a given data point, while y'_i is the corresponding predicted value. The terms $\bar{y}$ and $\bar{y}'$ represent the averages of the true and predicted values across all data points, respectively.

3 Results and discussions

3.1 Optimized predictions for compressive strength

The process of selecting model parameters was executed through the utilization of Bayesian optimization. A comprehensive dictionary of hyperparameters was curated for each model, encompassing specific ranges for each parameter. Subsequently, the iterative procedure ensued until the achievement of convergence in the R^2 value. Following this optimization endeavor aimed at maximizing metrics, a crucial fine-tuning phase was imperative to mitigate overfitting concerns that transpired with particular parameter configurations. In certain instances, bootstrap aggregating (bagging) was implemented to curb overfitting tendencies. Empirical refinements were implemented to ascertain the optimal configurations that strike a balance between overfitting and underfitting in both the training and testing datasets. Furthermore, in order to enhance the evaluation of the model's generalization capacity, particularly considering the constraint of a limited dataset size, the technique of K-fold cross-validation was employed. This approach serves to alleviate biases in data partitioning for training and testing by recurrently resampling and estimating model metrics. The K-fold cross-validation method involves dividing the dataset into multiple subsets for the purposes of training and testing, thereby furnishing a more precise and unbiased appraisal compared to a standard train/test division. By segmenting the dataset into ten uniformly sized folds, the model was trained on nine folds and evaluated on the remaining fold. This iterative process was reiterated ten times, with the validation subset rotating in each iteration, culminating in the computation of the final evaluation metric through the averaging of results. The adoption of the tenfold cross-validation methodology underscores the model's capacity for generalization and reliability concerning its performance (Kohavi, 1995).

For the 126-sample dataset, 80% of the data was allocated for the purpose of training and constructing the machine learning model, with the remaining 20% set aside for testing. Both the training and testing datasets were selected in a random manner to ensure fair representation. Table 3 presents the results for each model,

including R^2 , MAE, RMSE, and Pearson's correlation for both the training and testing sets.

According to Table 3, the XGBoost model achieved the highest R^2 scores, with 0.985 on the training set and 0.945 on the test set, paired with the lowest MAE (1.616 for training, 4.575 for testing) and RMSE (2.871 for training, 5.921 for testing). These results indicate robust predictive accuracy and minimal error in the model's estimations. Evidently, the AdaBoost algorithm also performed commendably, attaining an R^2 of 0.974 on the training data and 0.945 on the test data, combined with notably low MAE and RMSE values. In comparison, the GBR model closely trailed behind, achieving R^2 values of 0.952 for the training set and 0.940 for the testing set, accompanied by Pearson correlation coefficients of 0.976 and 0.971, respectively, revealing a strong positive association between predicted values and actual data points. Notably, the LightGBM Regressor displayed robust performance on the training dataset, garnering an R^2 of 0.953, yet exhibited a slightly diminished performance on the test set, with an R^2 of 0.926. Conversely, the RF and DT models exhibited slightly weaker performance, with R^2 scores falling below 0.93 for both the training and testing datasets, although they still demonstrated acceptable correlation outcomes. The Multi-Layer Perceptron (MLP) model delivered a solid performance, with an R^2 of 0.945 on the training set and 0.932 on the test set, thereby maintaining a favorable balance between error metrics and correlation. Overall, XGBoost and AdaBoost emerged as the top-performing algorithms, characterized by high predictive accuracy and superior generalization capabilities across the spectrum of evaluation metrics.

When comparing the performance of the three types of models—boosting models, tree-based models, and MLP—a notable pattern emerges, demonstrating that boosting models (including GBR, XGBoost, LightGBM, and AdaBoost) outperform the other two types in terms of accuracy and generalization. The boosting models exhibit minimal overfitting, as evidenced by the

narrow performance differentials between training and test datasets. Notably, XGBoost and AdaBoost consistently demonstrate high R^2 scores, low error metrics, and robust Pearson correlations, indicating their efficacy in handling both the training data and unseen information, thereby solidifying their position as the most resilient model variants in this particular context. In contrast, the tree-based models RF and DT display more conspicuous overfitting tendencies. Although these models perform admirably on the training data, their efficiency noticeably diminishes when tested on unseen datasets, as reflected in the reduced R^2 scores and elevated MAE and RMSE values. Notably, the Decision Tree model emerges as the most challenged in terms of generalization, showcasing the widest performance differential between training and testing outcomes, which renders it less suitable for accurate predictions on novel datasets. In the spectrum of model performance, the MLP model occupies an intermediate position between the boosting models and tree-based models. Although the MLP model does not attain the same pinnacle of accuracy as the boosting models, it displays superior generalization capabilities compared to the tree-based models, with a minimized performance gap between training and testing data. The MLP model exhibits limited overfitting tendencies and maintains robust correlation metrics; however, its error rates slightly exceed those of the top-performing boosting models. In summary, the analysis underscores that boosting models strike an optimal balance between accuracy and generalization, with the MLP model serving as a viable alternative that offers competitive performance. In contrast, the tree-based models are more susceptible to overfitting, thus emphasizing the inherent advantages of boosting models in achieving consistent and reliable predictive modeling outcomes.

Figure 4a–g display scatter plots of compressive strength predictions versus experimental values for the training and testing sets of the GBR, XGBoost, LightGBM, AdaBoost, RF, DT, and MLP models, respectively.

Table 3 Results of machine learning metrics for compressive strength

Boosting models	R^2		MAE		RMSE		Pearson's correlation	
	Train	Test	Train	Test	Train	Test	Train	Test
GBR	0.952	0.940	3.696	4.664	5.080	6.195	0.976	0.971
XGBoost	0.985	0.945	1.616	4.575	2.871	5.921	0.992	0.974
LightGBM	0.953	0.926	3.817	5.218	5.000	6.879	0.977	0.966
AdaBoost	0.974	0.945	2.397	3.742	3.742	5.941	0.987	0.973
RF	0.928	0.903	4.173	5.170	6.225	7.863	0.970	0.953
DT	0.928	0.893	4.433	5.746	6.201	8.273	0.973	0.954
MLP	0.945	0.932	4.379	4.884	5.416	6.578	0.984	0.967

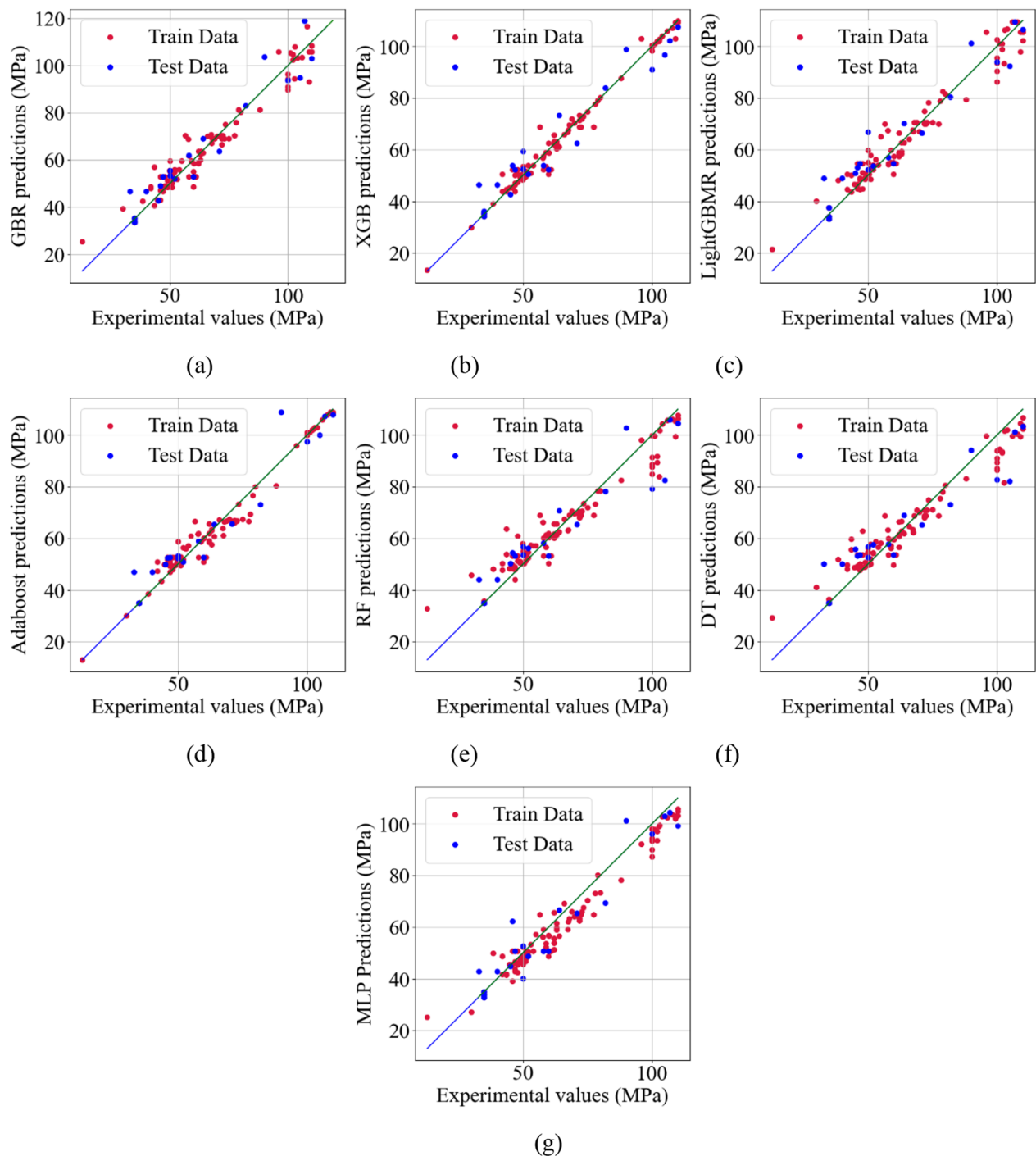


Fig. 4 Scatter plots of the experimental versus predicted values of compressive strength for **a** GBR, **b** XGBoost, **c** LightGBMR, **d** AdaBoost, **e** RF, **f** DT, and **g** MLP

The significant correlation observed between the predicted and measured compressive strength values not only underscores the robust predictive capacity of these models but also suggests a high level of performance.

The boosting models have exhibited superior performance, as shown in Fig. 5a–d, which present the results of tenfold cross-validation for the R^2 , RMSE, and MAE metrics across individual folds for GBR, XGBoost, LightGBM, and AdaBoost, respectively. Notably, there is a

notable consistency in R^2 values with low variance across all models. While the MAE and RMSE errors demonstrate less than 30% variation for most models, XGBoost stands out with the lowest errors but with a higher variability in MAE (49.88%) and RMSE (53.91%). In terms of the average cross-validation metrics for predicting compressive strength, the GBR model yields an R^2 value of 0.934 on average, with an MAE of 3.696 and RMSE of 4.959. The XGBoost model shows an average R^2 of 0.984, an MAE of 1.616, and an RMSE of 2.527. The LightGBM model achieves an average R^2 of 0.947, an MAE of 3.817, and an RMSE of 4.832, while the AdaBoost model attains an average R^2 of 0.968, an MAE of 2.397, and an RMSE of 3.684. These findings indicate that the boosting models offer stability, reliability, and resistance to overfitting, thereby demonstrating their proficiency in delivering precise predictions with minimal variance. Additionally, these models exhibit promise for generalizing to new, unseen data, which enhances their suitability for practical applications outside the confines of the training dataset.

In order to evaluate the robustness of the model in the face of input perturbations, Gaussian noise was deliberately introduced into the test dataset at varying intensity levels. Specifically, random values were sampled from a normal distribution and added to the test features at noise magnitudes of 5%, 10%, 15%, and 20%. A comparative analysis of the model's performance, as measured by the R^2 metric, was conducted between the original test dataset and the perturbed test datasets to gauge its resilience in the presence of noisy inputs. The outcomes of this evaluation are tabulated in Table 4 for the GBR, XGBoost, LightGBM, and AdaBoost models, respectively. This analysis quantifies the model's ability to generalize in the face of input variability, thus providing insights into its robustness in real-world applications where noise or uncertainty may arise.

Across all noise levels, LightGBM Regressor was the most sensitive to noise. Specifically, the R^2 exhibited a steady decline of 9.535% at a noise level of 5%, deteriorating further to a 13.319% decrease at a noise level of 20%.

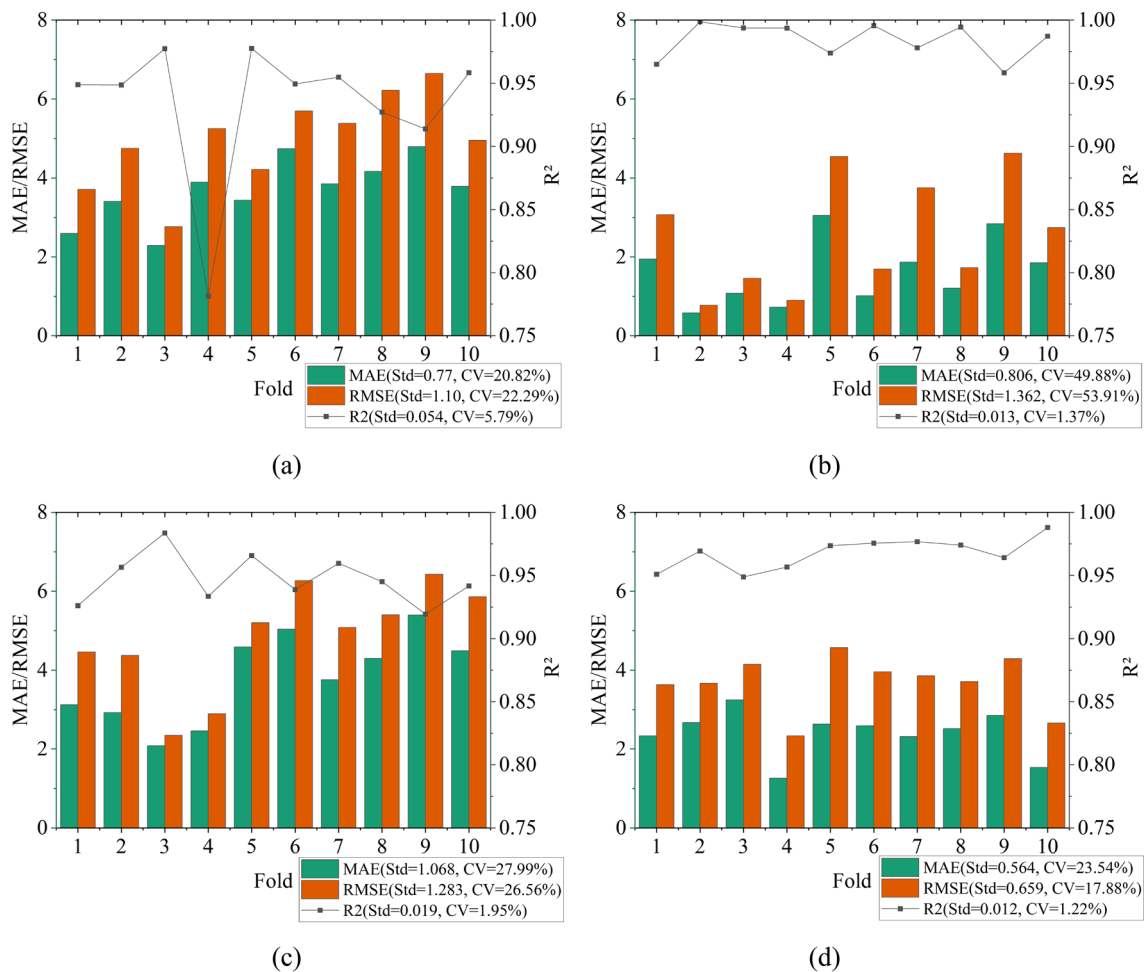


Fig. 5 Results of each fold for the tenfold cross-validation approach for compressive strength for **a** GBR, **b** XGBoost, **c** LightGBM, and **d** AdaBoost

Table 4 Percentage changes for R^2 metric with the addition of noise for models predicting compressive strength

Noise (%)	GBR (%)	XGBoost (%)	LightGBMR (%)	AdaBoost (%)
5	0.000	0.724	9.535	0.154
10	0.003	0.834	9.544	0.163
15	0.545	1.151	12.483	0.292
20	0.596	4.435	13.319	0.302

XGBoost also displayed escalating sensitivity to noise as the levels increased. This was evidenced by a noticeable escalation from a reduction of 1.151% at a noise level of 15% to a more pronounced decrease of 4.435% at a noise level of 20%, highlighting a significant impact on model performance that accelerated with higher levels of noise.

In contrast, GBR and AdaBoost presented marginal fluctuations in R^2 with the introduction of noise. Specifically, GBR showcased a R^2 variation of less than 1% across all noise levels, indicating a robust resistance to input variability. Similarly, AdaBoost displayed a comparable pattern, with a noteworthy impact observed at a 20% noise level resulting in a mere 0.302% change, underscoring its capability to uphold consistent performance in the presence of noise. In summary, these findings suggest that while LightGBM and, to a lesser extent, XGBoost exhibit greater susceptibility to input disturbances, GBR and AdaBoost exemplify a heightened level of resilience, maintaining a stable performance profile across varying noise intensities.

3.2 Optimized predictions for pump-speed

In accordance with the established procedure for assessing compressive strength, a comparable approach utilizing a common dictionary entailing specific values for each model was employed. Subsequent to this, an iterative process was initiated, coupled with a fine-tuning of the model in question. Empirical adjustments were carried out in order to determine the optimal configurations

that strike a delicate balance between the risks of overfitting and underfitting.

For the 126-sample dataset, 80% of the data was dedicated to the training and development of the machine learning model, while the remaining 20% was earmarked for testing purposes. The selection of training and testing data was randomized to ensure impartiality. The following Table 5 elucidates the outcomes in terms of R^2 , MAE, RMSE, and Pearson's correlation for both the training and testing sets regarding pump speed. The findings reveal that among the various machine learning models employed for pump speed predictions, boosting models, particularly XGBoost and AdaBoost, exhibited superior accuracy and generalization capabilities. XGBoost demonstrated the most outstanding performance, achieving an R^2 value of 0.953 on the test set, in conjunction with the lowest MAE and RMSE metrics, hence underscoring its exceptional predictive accuracy. Noteworthy is the commendable performance of AdaBoost, which closely trailed behind by maintaining high R^2 and Pearson correlation coefficients, all while displaying minimal performance degradation between the training and test sets. While GBR and LightGBMR also boasted commendable performances, they displayed marginally higher error rates, indicative of less precision in comparison to XGBoost and AdaBoost. Conversely, tree-based models like Random Forest and Decision Tree exhibited signs of overfitting, with Decision Tree, in particular, showcasing the highest error rates and weakest generalization capacity. The MLP model demonstrated respectable performance, boasting satisfactory generalization and relatively lower error metrics, albeit falling short in precision as opposed to the boosting models. The results of this analysis underline the effectiveness of boosting models for pump speed prediction, with XGBoost and AdaBoost emerging as the most proficient models in this context.

Figure 6a–g show the scatter plots of pump-speed predictions versus experimental values for the training and testing sets for GBR, XGBoost, LightGBMR, AdaBoost,

Table 5 Results of machine learning metrics for pump-speed

Boosting models	R^2		MAE		RMSE		Pearson's correlation	
	Train	Test	Train	Test	Train	Test	Train	Test
GBR	0.952	0.938	8.078	8.281	10.725	11.788	0.977	0.976
XGBoost	0.988	0.953	3.499	6.007	5.335	10.265	0.994	0.985
LightGBMR	0.942	0.920	8.899	8.494	11.843	13.392	0.971	0.968
AdaBoost	0.985	0.948	3.308	7.260	6.000	10.830	0.993	0.988
RF	0.977	0.911	4.525	8.042	7.463	14.141	0.991	0.980
DT	0.927	0.884	9.058	11.289	13.282	16.160	0.978	0.976
MLP	0.949	0.940	6.625	7.499	11.069	11.632	0.976	0.974

RF, DT, and MLP models, respectively. The strong alignment between the predicted and actual pump speeds demonstrates the high predictive accuracy and performance of these models.

Upon conducting 10-fold cross-validation on the boosting models, denoted as pump-speed output, the outcomes for the coefficients of R^2 , RMSE, and MAE metrics for GBR, XGBoost, LightGBM, and AdaBoost are visually represented in Fig. 7a–d, respectively. The

consistency of R^2 for the models with low variation is noticeable in the case of GBR and LightGBM. However, XGBoost and AdaBoost display greater variability, with values ranging from 37% to approximately 47%. The MAE and RMSE errors are markedly lower in the XGBoost and AdaBoost models in comparison to GBR and LightGBM. In terms of average metrics obtained through cross-validation for pump-speed prediction, the GBR model demonstrates an average R^2 value of 0.932, an

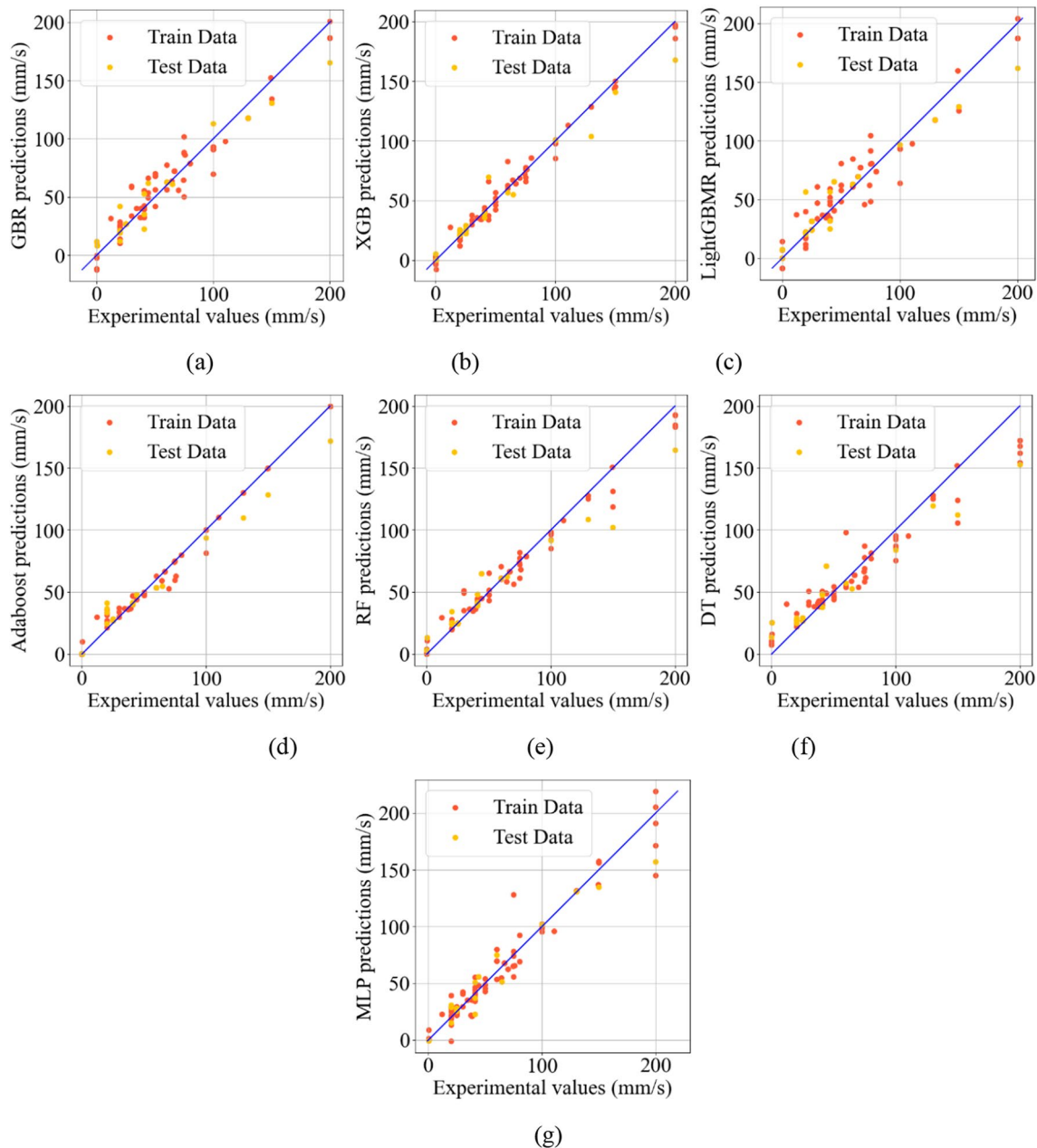


Fig. 6 Scatter plots of the experimental versus predicted values of pump-speed for **a** GBR, **b** XGBoost, **c** LightGBM, **d** AdaBoost, **e** RF, **f** DT, and **g** MLP

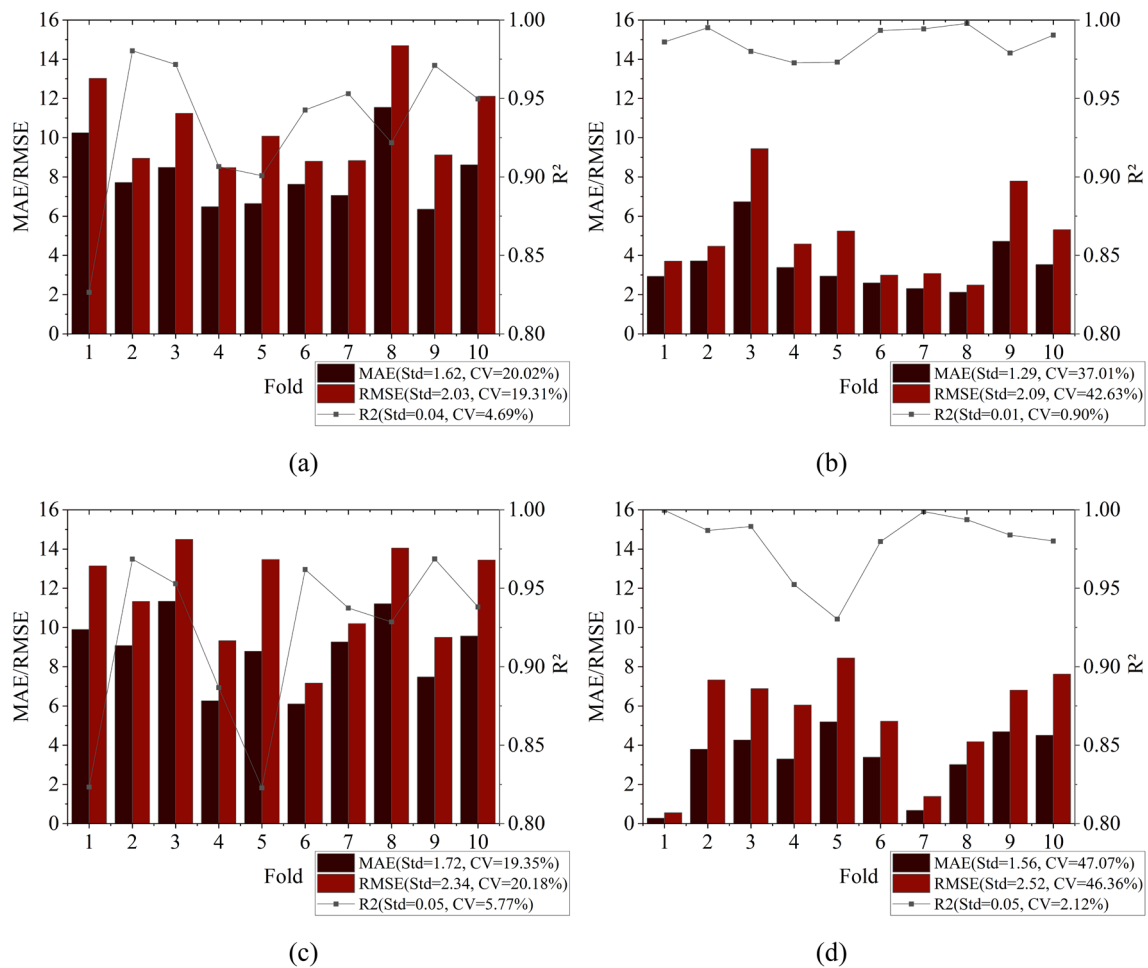


Fig. 7 The results of each fold for the tenfold CV approach for pump-speed for **a** GBR, **b** XGBoost, **c** LightGBM, and **d** AdaBoost

MAE of 8.078, and an RMSE of 10.531. In contrast, the XGBoost model yields an average R^2 of 0.986, with corresponding MAE and RMSE values of 3.499 and 4.908, respectively. The LightGBM model displays an average R^2 of 0.919, an MAE of 8.899, and an RMSE of 11.609, while the AdaBoost model shows an average R^2 of 0.980, an MAE of 3.308, and an RMSE of 5.444. These results suggest that, overall, the boosting models exhibit stability, reliability, and a capacity to avoid overfitting, thereby demonstrating their ability to provide precise predictions with reduced variance. Furthermore, the models showcase potential for generalization to novel, unseen data, rendering them suitable for practical applications beyond the confines of the training dataset.

Table 6 illustrates the impact of noise on the percentage changes of R^2 in models utilized to predict pump speed. The analysis reveals varying degrees of resilience across different machine learning algorithms, namely GBR, XGBoost, LightGBM, and AdaBoost. GBR demonstrates a remarkable degree of stability, with minimal

variation in R^2 values up to a noise level of 10%, showing only marginal increases at higher noise levels (0.344% at 15% and 0.733% at 20%). This pattern suggests a high level of robustness and resilience in GBR against noise interference. On the other hand, XGBoost exhibits slightly higher sensitivity to noise, with R^2 fluctuations ranging from 0.027% at 5% noise to 0.880% at 20% noise. Despite this, the changes remain relatively modest, indicating a moderate level of robustness in XGBoost. In

Table 6 Percentage changes for R^2 metric with the addition of noise for models predicting pump-speed

Noise (%)	GBR (%)	XGBoost (%)	LightGBM (%)	AdaBoost (%)
5	0.000	0.027	3.023	0.152
10	0.000	0.202	6.546	0.716
15	0.344	0.759	7.018	0.931
20	0.733	0.880	27.465	1.503

contrast, LightGBM experiences a significant decline in performance, particularly evident at a noise level of 20% where the R^2 change reaches a substantial 27.465%. This marked increase highlights LightGBM's vulnerability to input noise, emphasizing the importance of considering this susceptibility in environments characterized by high levels of noise. AdaBoost, on the other hand, demonstrates incremental sensitivity to noise, with R^2 changes gradually escalating from 0.152% at 5% noise to 1.503% at 20%, showcasing a consistent and controlled response to noise interference. Overall, GBR is the most robust model for predicting pump speed under noisy conditions, followed by XGBoost and AdaBoost. In contrast, LightGBM displays pronounced sensitivity to noise, particularly at elevated noise levels.

3.3 Optimized predictions for carbon footprint

The process of selecting boosting model parameters for estimating carbon footprint followed the same systematic approach and utilized the identical range of dictionary values as applied to the other variables. Similar to the other parameters, empirical fine-tuning was conducted to ascertain the optimal configurations that strike a balance between overfitting and underfitting.

An outlier was identified in the carbon footprint data, and subsequent preprocessing led to the utilization of a dataset comprising 125 samples. Following the preprocessing stage, 80% of the data was dedicated to training and constructing the machine learning model, while the remaining 20% was set aside for the testing set. The results across all models are presented in Table 7, showcasing the R^2 , MAE, RMSE, and Pearson's correlation outcomes for both the training and testing datasets. The results of predicting carbon footprints using various machine learning models exhibit exceptional precision, as evidenced by the models achieving nearly perfect R^2 and Pearson's correlation values, underscoring their robust predictive capabilities. Notably, XGBoost and AdaBoost stand out as frontrunners, producing predictions of

almost flawless accuracy with minimal errors, positioning them as the top-performing models. The Gradient Boosting Regressor (GBR) and Random Forest models also demonstrate strong performance albeit with slightly elevated error metrics. Conversely, LightGBM and MLP showcase commendable accuracy but with greater prediction errors in comparison to the leading models. Surprisingly, the Decision Tree model performs on par with XGBoost and AdaBoost, matching their exemplary predictive accuracy. Overall, the boosting models, particularly XGBoost and AdaBoost, provide the most precise and reliable predictions for carbon footprint.

Figure 8a–g show the scatter plots of carbon footprint predictions versus experimental values for the training and testing sets of the GBR, XGBoost, LightGBM, and AdaBoost models, respectively. The significant alignment observed between the predicted and observed carbon footprint values provides substantial support for the exceptional predictive proficiency of these models.

The outcomes derived from the 10-Fold cross validation analysis concerning the carbon footprint output are eloquently illustrated in Fig. 9, segmented into individual plots denoted as a–d. These plots meticulously reveal the statistical performance pertaining to the R^2 , RMSE, and MAE metrics throughout each fold for GBR, XGBoost, LightGBM, and AdaBoost, respectively. Across the diverse models scrutinized, a consistent observation emerges with regards to the R^2 values, displaying commendable levels of reliability across all models. However, it is discerned that GBR manifests notable discrepancies in predictive accuracy, marked by heightened error rates, whereas XGBoost accentuates exceedingly elevated errors. Conversely, LightGBM and AdaBoost portray a more stable performance, characterized by relatively diminished fluctuations in error metrics throughout the cross-validation process. Upon elucidating the average metric results for carbon footprint prediction, the GBR model presents an average R^2 value of 0.999, an MAE of 7.110, and an RMSE of 11.128. For the XGBoost model,

Table 7 Results of machine learning metrics for carbon footprint

Boosting models	R^2		MAE		RMSE		Pearson's correlation	
	Train	Test	Train	Test	Train	Test	Train	Test
GBR	0.999	0.998	7.110	11.587	12.861	23.750	1.000	0.999
XGBoost	1.000	0.999	3.766	10.643	10.835	19.748	1.000	0.999
LightGBM	0.997	0.994	18.564	26.501	29.899	43.840	0.999	0.997
AdaBoost	1.000	0.999	5.451	10.575	10.240	20.923	1.000	0.999
RF	0.999	0.998	7.110	11.587	12.861	23.750	1.000	0.999
DT	1.000	0.999	3.766	10.643	10.835	19.748	1.000	0.999
MLP	0.997	0.994	18.564	26.501	29.899	43.840	0.999	0.997

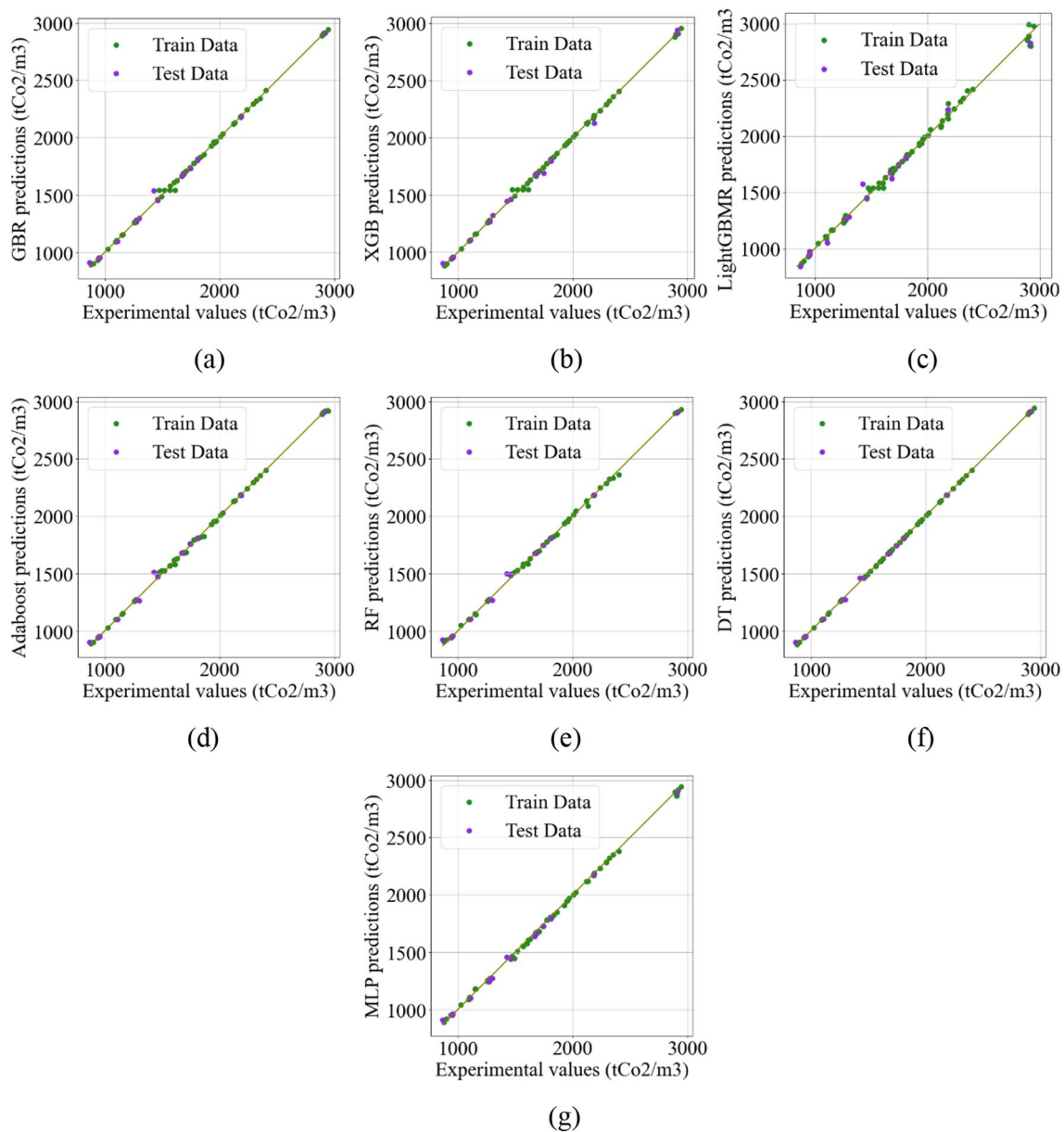


Fig. 8 Scatter plots of the experimental versus predicted values of carbon footprint for **a** GBR, **b** XGBoost, **c** LightGBMR, **d** AdaBoost, **e** RF, **f** DT, and **g** MLP

the average R^2 is 0.999, MAE is 3.766, and RMSE is 7.931. For the LightGBMR model, the average R^2 is 0.997, MAE is 18.564, and RMSE is 27.102. For the AdaBoost model, the average R^2 is 0.999, MAE is 5.451, and RMSE is 9.606. Noteworthy is the fact that XGBoost manages to achieve the lowest predictive error rates; nonetheless, it concurrently registers the highest variance in both MAE and RMSE metrics. AdaBoost proves to be the most proficient model overall, characterized by diminished error rates alongside a moderate coefficient of variability. GBR

demonstrates an intermediary level of predictive accuracy along with a moderate variability coefficient, while LightGBM displays a lower variability coefficient yet occasionally records the most substantial errors across folds when juxtaposed with other models. These compelling findings underscore the inherent stability and reliability of the analyzed boosting models in deciphering and generalizing unseen data pertaining to the carbon footprint of mixtures, albeit with a notable exception

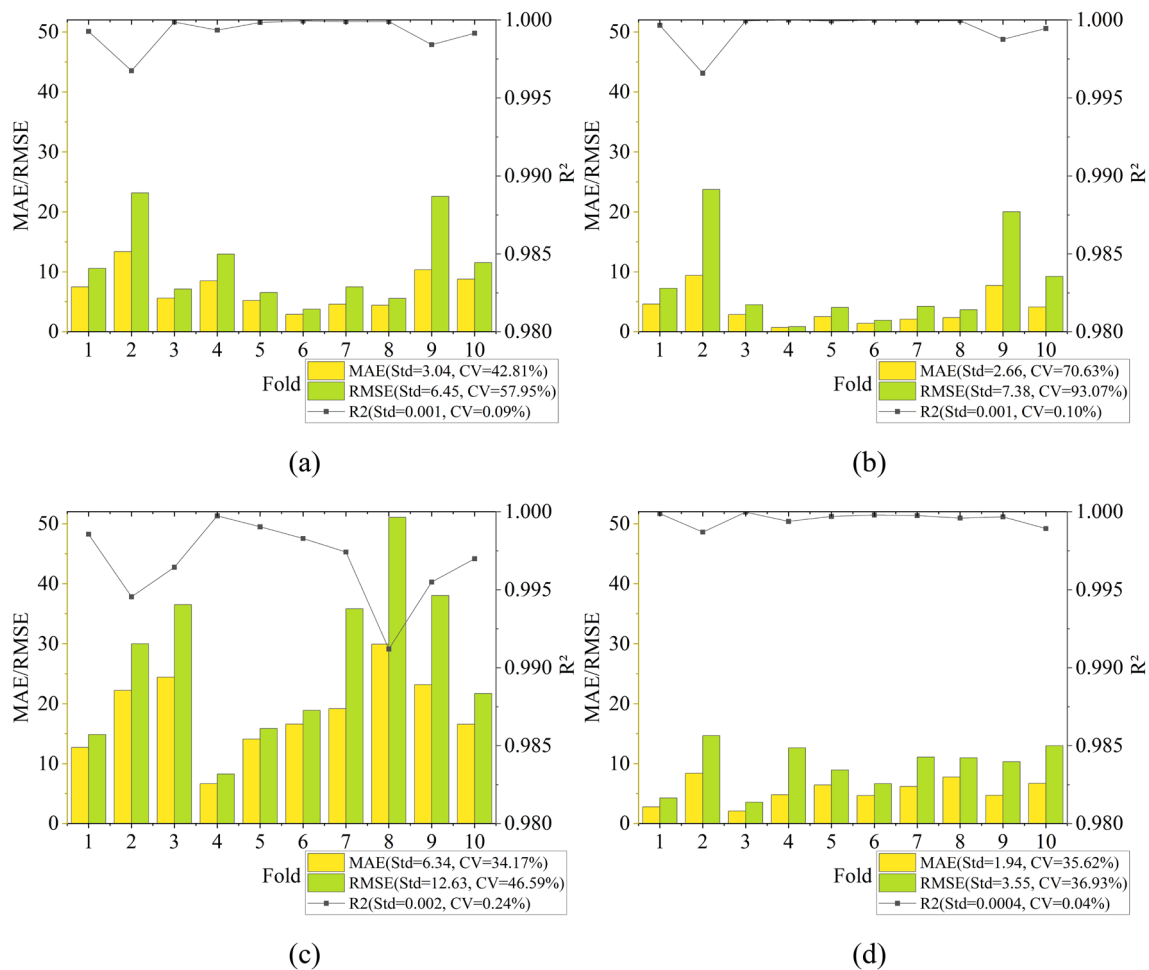


Fig. 9 The results of each fold for the tenfold CV approach for carbon footprint for **a** GBR, **b** XGBoost, **c** LightGBM, and **d** AdaBoost

for XGBoost which stands out due to its pronounced variability.

Table 8 illustrates the impact of noise on the variations in the R^2 percentage changes for models predicting carbon footprint. Both GBR and AdaBoost demonstrate exceptional resilience, with minimal R^2 percentage changes across all noise levels. They demonstrate stability even under the influence of 20% noise, with changes as minor as 0.024% and 0.0014%, respectively. On the other hand, XGBoost displays moderate sensitivity, as evidenced by the increase in R^2 variations from 0.001% at 5% noise to 0.823% at 20%, implying a fair degree of robustness alongside a slight vulnerability to higher noise levels. Conversely, LightGBM is noticeably impacted by noise, showing notable shifts in R^2 percentages from 3.143% at 5% noise to 5.523% at 20%. This observation suggests that LightGBM may be less appropriate for datasets characterized by high levels of noise. In summary, the findings mirror those of prior analyses, highlighting GBR and AdaBoost as the

Table 8 Percentage changes for R^2 metric with the addition of noise for models predicting carbon footprint

Noise (%)	GBR (%)	XGBoost (%)	LightGBM (%)	AdaBoost (%)
5	0.000	0.001	3.143	0.0009
10	0.003	0.383	3.973	0.0005
15	0.024	0.802	5.483	0.0007
20	0.024	0.823	5.523	0.0014

most resilient models for predicting carbon footprints. Meanwhile, the sensitivity of LightGBM underscores the necessity for cautious evaluation when operating in noisy data environments.

4 Sensitivity analysis using SHAP interpretation

4.1 Feature importance with positive and negative effect for outputs

To enhance the interpretability of boosting models, the SHAP framework, particularly the “shap.Explainer,” was employed, given its extensively documented efficacy (Jabeur et al., 2021; Nazar et al., 2023; Ullah et al., 2023). In order to examine the positive and negative impacts, the AdaBoost model was chosen as it exhibited robust predictive capabilities across all three outcomes, attaining heightened accuracy with minimal error metrics while preserving a balanced coefficient of variation. Summary plots displayed in Fig. 10a–c amalgamate the significance of variables with their respective effects in the AdaBoost model pertaining to CS, PS, and CF, respectively. These visualizations elucidate whether the influence exerted by the predictors is positive or negative. Each data point within the plots corresponds to a SHAP value attributed to a particular variable, which, when viewed collectively, enables the interpretation of the predictors’ general behavior. A color gradient ranging from blue through purple to red is utilized to signify whether the input value contributes to low, moderate, or high predictions, respectively. Instances where there is a concentration of SHAP values result in vertically stacked dots. For example, in Fig. 10a, the top-right red dot within the

admixture row represents a SHAP value approaching 40, indicating the potential to elevate CS by over 30 units from the mean value.

In the analysis of Fig. 10a pertaining to the compressive strength output, it is discernible that high concentrations of admixture manifest positive SHAP values as evidenced by the crimson data points extending towards the right in the inaugural row, reaching SHAP values of approximately 35. Conversely, a substantial portion of instances representing lower concentrations of admixture on the plot exhibit markedly negative SHAP values, thereby shifting the data points to the leftward side. Noteworthy are certain violet data points that exhibit a substantial reduction in SHAP value, displaying a negative value hovering around -18. This pattern signifies a direct correlation between heightened levels of admixture and elevated predicted compressive strength, while lower and moderate concentrations tend to correspond with lower predicted compressive strength values. Analogous trends are observed in the plots for fine aggregate and silica fume features, albeit with a less pronounced effect, with SHAP values ranging from -12 to a peak of approximately 20 for fine aggregate, and from -5 to 15 for silica fume. In the instance of SP, owing to the distinct variations in types, the plot illustrates that both high and low concentrations exhibit varying effects on compressive strength,

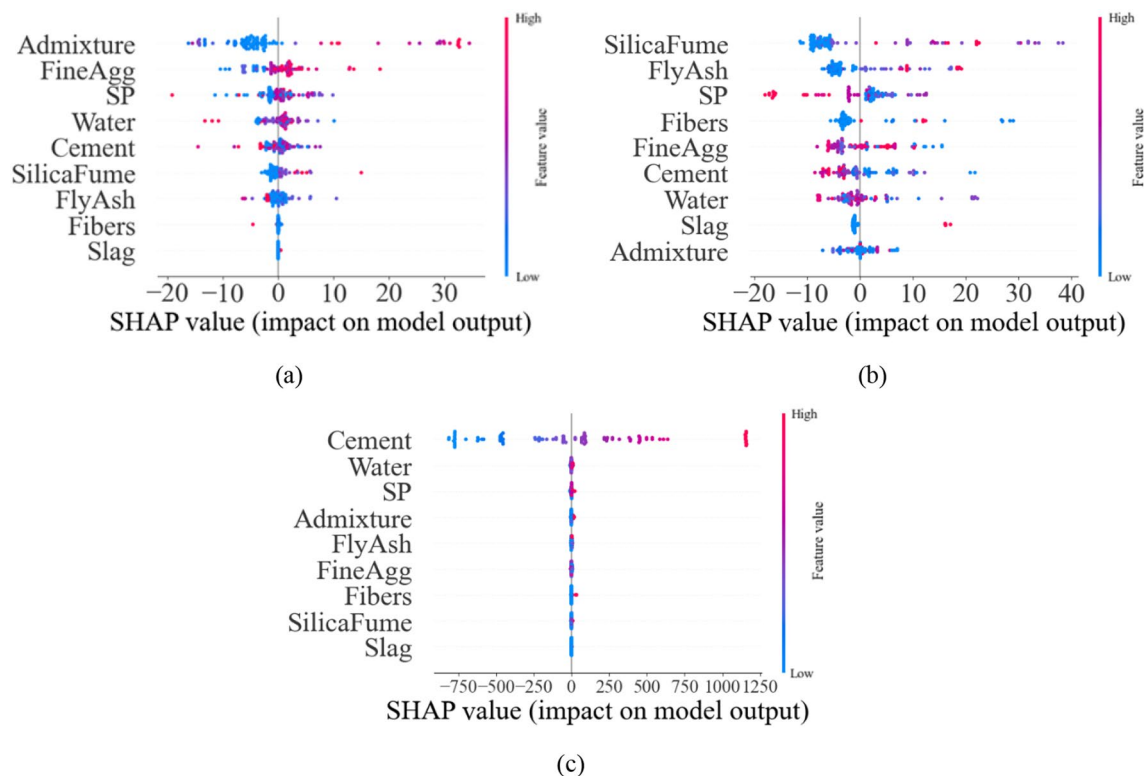


Fig. 10 SHAP summary plot for 3DCP combining feature importance with feature effects on AdaBoost models for **a** CS, **b** PS, and **c** CF

indicating both negative and positive influences. With regards to water, the influence is ambiguous, encompassing both positive and negative impacts. Noteworthy is the observation that instances characterized by high water concentrations often lead to a marked reduction in compressive strength, as reflected by data points extending to the left with SHAP values converging at around -15. Conversely, certain azure data points elicit a positive influence on compressive strength, conversely reflected by SHAP values of approximately 10. Turning to cement and fly ash, it is apparent that elevated concentrations have a tendency to adversely affect compressive strength, while lower and moderate levels typically yield a positive influence. In the case of fibers and slag, the impact is relatively minimal, although a singular crimson instance in the plot implies that heightened quantities of fibers negatively affect compressive strength, whereas lower concentrations exert a positive influence. Conversely, for slag, an opposing trend is observed wherein a single instance suggests that elevated levels correlate with a positive effect on compressive strength, while lower concentrations correspond with diminished compressive strength.

The present analysis, grounded in SHAP methodology, offers a comprehensive evaluation of specific material modifications that can enhance the compressive strength of 3DCP mixtures while concurrently considering other critical performance attributes. Augmenting the content of admixtures consistently manifests as a fruitful approach that significantly bolsters compressive strength, thereby holding promise for reinforcing the structural integrity of load-bearing constructions. Furthermore, the positive influence of fine aggregates on compressive strength indicates that moderate increments in their proportion can amplify strength without compromising the overall workability of the mixture. Conversely, the presence of elevated water levels, as inferred from the observation of adverse SHAP values, has a detrimental effect on compressive strength, underscoring the imperative of maintaining water content within an optimal range to uphold the dual imperatives of strength and workability. The nuanced insights provided by the SHAP analysis reveal potential trade-offs inherent in optimizing concrete mixtures. It becomes evident that while augmenting the admixture content has a positive impact on compressive strength, it may feasibly influence other critical parameters such as setting time and curing processes. Similarly, reducing water content can enhance compressive strength but poses a potential risk to pumpability, a vital consideration for the efficacious extrusion process in 3D printing applications. High levels of fly ash and cement may negatively impact CS but could improve durability or workability.

For Fig. 10b, depicting varying pump speeds in relation to the presence of silica fume, fly ash, fibers, slag, SP, fine aggregate, cement, water, and admixture, a consistent trend is observed among certain materials. Silica fume, fly ash, fibers, and slag tend to demonstrate analogous patterns. The data points highlighted in blue, denoting lower levels of these materials, exhibit negative SHAP values correlating with a detrimental impact on pump speed. Conversely, when these materials are present in higher quantities, indicated by green points with positive SHAP values, their influence on pump speed predictions is positive. In contrast, the presence of SP, fine aggregate, and cement in reduced amounts appears to have a positive effect on pump speed, while higher levels of these materials tend to have a negative impact. The behavior of water and admixture introduces a degree of complexity to this analysis. Both these components exhibit a mixed pattern, with varying quantities affecting pump speed predictions in both positive and negative directions.

This scholarly examination, utilizing the SHAP methodology, delves into the relationship between pump-speed in the context of 3D concrete printing, aiming to enhance flow characteristics and operational efficiency. The study reveals that augmenting the proportions of silica fume, fly ash, fibers, and slag can significantly enhance pump speed by mitigating flow resistance, thereby facilitating a smoother and expedited pumping process. Conversely, decreasing the levels of SP, fine aggregate, and cement also has a positive impact on pump speed, as lower concentrations help to prevent excessive viscosity that might impede the flow dynamics. The influence of water and admixture levels exhibits varied effects on pump speed, necessitating a delicate balance with other constituents to ensure optimum performance. However, it is important to note that these adjustments entail nuanced trade-offs. While higher percentages of slag and silica fume can enhance pumpability, there is a potential tradeoff in terms of prolonged curing times or diminished early strength, which could pose challenges in time-sensitive construction projects. Likewise, reducing the amounts of SP, fine aggregate, and cement may boost pump speed, yet it runs the risk of compromising the cohesion of the mixture, thereby increasing the likelihood of segregation or compromising structural integrity. Therefore, any modifications to water and admixture quantities must be executed with caution, as drastic alterations could impact durability and homogeneity of the concrete mixture.

The analysis of Fig. 10c reveals intriguing insights into the predictions of carbon footprint within the context of varying quantities of essential construction materials. Notably, a substantial quantity of cement is linked to heightened carbon footprint predictions, with SHAP

values exceeding 1000 for positive impact and descending to around -800 for negative impact. Furthermore, the impact of water and admixture on carbon footprint forecasts aligns with a comparable trend, albeit on a more modest scale. Within this framework, the speed of the pumping process emerges as a significant factor influencing carbon footprint outcomes, contingent upon the energy efficiency of operations. Optimal pump speeds, characterized by higher velocities and enhanced efficiency, are associated with reduced carbon footprint projections. In contrast, lower to medium velocities often contribute to heightened carbon footprint estimates due to prolonged processing durations. However, intriguingly, the analysis also unveils instances where low pump speeds, depicted as a vertical line with blue data points, exhibit the potential to mitigate carbon footprint values under specific conditions, such as varying extrusion rates and external environmental factors. For SP, their diverse types manifest varied influences on carbon footprint estimations, exerting both positive and negative effects depending on their inherent characteristics. Fly ash, when present in moderate quantities, demonstrates a tendency to diminish SHAP values, thereby curbing carbon footprint projections. Conversely, scenarios involving either low or high abundance of fly ash tend to elevate SHAP values and correspondingly increase carbon footprint estimations. Moreover, the remaining features exhibit a blend of effects on carbon footprint outcomes, albeit with relatively lesser degrees of influence compared to the prominent factors discussed above.

This analysis based on SHAP for carbon footprint in 3D concrete printing reveals strategies for mitigating emissions without sacrificing performance standards. Substituting a portion of cement with fly ash at moderate levels shows promise, as evidenced by the negative SHAP values associated with medium fly ash concentrations, facilitating a reduction in carbon intensity without compromising structural integrity. Careful management of water and admixture quantities is also pivotal, as excessive usage escalates the carbon footprint; optimal levels of these components can be adjusted to further diminish emissions while preserving workability. Noteworthy is the crucial influence of pump speed on operational efficacy, with higher speeds correlating with lowered carbon footprint attributed to reduced pumping duration and energy consumption. This suggests that optimizing pump speed for energy efficiency can help minimize emissions. Nevertheless, these adjustments engender potential trade-offs. For instance, diminishing cement content may compromise strength, while overly reducing water or admixture levels could impede flowability, an essential aspect of the extrusion process. Similarly, although elevated pump speeds offer energy efficiency benefits, they

may compromise the precision of printing, particularly in projects necessitating meticulous control over detailing.

5 Multi-objective optimization

5.1 Tri-objective optimization

In this segment, a tri-objective optimization is performed aiming to maximize compressive strength, and simultaneously minimize both cost and carbon footprint, by employing the renowned AdaBoost model. The DEAP library was utilized in formulating the multi-objective optimization conundrum. Various components of the genetic algorithm were meticulously configured, encompassing the generation of random integer attributes, as well as individual and population initialization. The genetic operators, including crossover, mutation, and selection, were finely tuned for optimal performance. The assessment of the genetic algorithm's efficacy involves scrutinizing the population's evolutionary trajectory across successive generations, with a keen focus on the stabilization of the Pareto front post a specific number of iterations.

Following sensitivity analysis, the evolutionary algorithm was executed over a span of 50 generations, with a populace comprising 100 unique individuals and an offspring size of 200, progressively refining the population and selecting individuals predicated on their fitness levels. Figure 11 portrays a 3D scatter plot depicting compressive strength juxtaposed with price and carbon footprint. Moreover, the Pareto front solutions are delineated utilizing a gradient color spectrum, wherein lighter pink hues connote inferior TOPSIS scores, indicative of solutions further from the optimal point, while deeper red shades denote superior TOPSIS scores, reflecting solutions in close proximity to the optimal point. The solution attaining the optimum TOPSIS score is accentuated as the paramount solution, adept at balancing the inherent trade-offs. Furthermore, the most cost-efficient solution is duly highlighted within the graphical representation.

Table 9 illustrates the combinations chosen through Pareto front analysis alongside their respective indices. The optimal blend, identified through TOPSIS for achieving a harmonized equilibrium among the specified objectives, and the most economically efficient combination are explicitly emphasized in the table.

The utilization of a multi-objective optimization approach proves to be instrumental for project managers and engineers in achieving a harmonious equilibrium among factors such as compressive strength, cost, and sustainability, all of which are pivotal in practical applications within the real-world scenario. A concrete illustration of this is evident in public infrastructure endeavors that aim to secure green building certification. Here, the

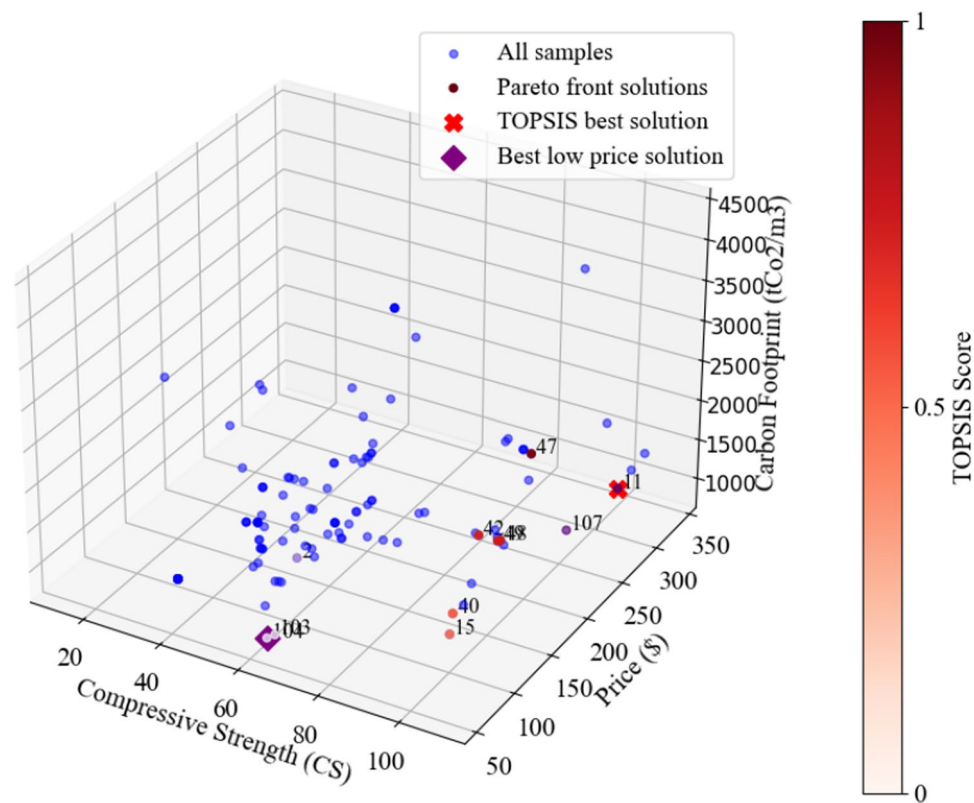


Fig. 11 Tri-objective optimization for price vs compressive strength vs carbon footprint

Table 9 Mixtures selected from Pareto front analysis for Tri-objective optimization

Pareto front solutions												
Index	C	W	FineAgg	FA	SF	SP	S	Admix	F	CS	Price	CF
11*	483	182	1074	0	268	10.7	322	6.4	19.6	109.0	280.2	1453.6
47	1000	375	1250	0	0	5	0	14	0	108.3	159.2	2912.7
48	750	300	1000	0	0	5	0	9	0	108.0	121.6	2184.5
49	750	250	1000	0	0	5	0	9	0	107.6	120.4	2184.5
42	750	250	750	0	0	5	0	9	0	103.0	120.3	2184.5
107	391	343.1	1260	213	213	16.3	0	0	0	102.7	243.8	1147.5
40	500	150	500	0	0	5.2	0	8	0	102.0	93.0	1455.9
15	376	150	1612	107	54	0	0	0	0	100.0	100.0	1101.0
103	323.5	303.2	817.3	0	0	2	0	0.2	0	64.9	56.9	939.7
104**	301.8	225.3	970.4	0	0	2	0	0.2	0	63.1	56.0	879.2
2	290	285	1211	278	145	0	0	7	0	52.0	148.3	873.9

* Best TOPSIS solution

** Best low-price solution

meticulous selection of a concrete mixture that satisfies structural performance and financial prerequisites while concurrently minimizing the carbon footprint emerges as a paramount factor, thereby aiding in adhering to stringent environmental regulations and sustainability targets. The genetic algorithm serves as an efficient tool in

navigating the intricate trade-offs existing among these various objectives, allowing decision-makers to consider mixtures that offer the desired compressive strength without exceeding budget constraints. The Pareto front solutions provide multiple optimal options, each presenting a distinct balance across these criteria. By applying

the TOPSIS ranking method, project teams can prioritize mixtures that best align with their ideal performance in all three dimensions—compressive strength, cost, and environmental impact. By adopting such an approach, not only do projects exhibit cost-effectiveness and structural robustness, but they also demonstrate adherence to sustainability objectives, thus contributing significantly to the cultivation of a conscious and adaptable construction sector.

6 Conclusions

The present research delved into an examination of variables pertaining to the compressive strength, pump rate, and carbon footprint in the context of 3D concrete printing (3DCP). Various boosting models were utilized alongside Bayesian optimization and fine-tuning techniques to bolster overall performance. The AdaBoost model showcased robust predictive capabilities, with XGBoost identified as somewhat less steadfast in forecasting carbon footprint implications. The performance of the models was assessed through key metrics, encompassing R^2 , RMSE, MAE, and Pearson's correlation, ultimately navigating the delicate equilibrium between excessive model complexity and oversimplification. Furthermore, the amalgamation of tenfold cross-validation methodologies with Gaussian noise injection corroborated the models' aptitude for generalization across novel datasets, underscoring their applicability in broader contexts, with tailoring considerations for XGBoost in select scenarios.

An in-depth SHAP analysis shed light on the salience of distinct features, presenting avenues for targeted material modifications to optimize the composition of 3DCP formulations. This interpretive framework advocates for a synergistic, sustainable approach to 3DCP, steering advancements in structural integrity, workability, and environmental sustainability.

Tri-objective optimization further aided practical decision-making by balancing compressive strength, cost, and environmental impact through the AdaBoost model, genetic algorithms, and TOPSIS ranking. This approach provides project managers with an array of optimal solutions aligning with performance benchmarks, financial constraints, and green imperatives.

The findings of this study bear concrete implications for the advancement of 3DCP technologies and the mitigation of CO₂ emissions. The methodologies devised herein stand as potent assets in the quest to devise novel 3DCP blends meeting predefined specifications and sustainability objectives. By integrating boosting models, SHAP insights, and multi-objective optimization strategies, this approach aids in the identification of pivotal parameters and the fine-tuning of mixture proportions to

enhance quality while curbing errors, thereby facilitating eco-friendlier construction practices.

The predictive models derived in this study evinced remarkable accuracy in prognosticating key properties of 3DCP, encompassing compressive strength, pump rate, and carbon footprint considerations. The validation of the models was predicated upon the segregation of testing datasets from their training counterparts, with both datasets derived from controlled laboratory experiments. The exigencies of practical limitations, inclusive of temporal constraints, finite resources, and material accessibility, precluded direct experimental validation in this instance. Nonetheless, a rigorous regime of cross-validation and resilience testing, supplemented by noise susceptibility analyses, underpinned the resilience of the models in diverse contexts. Looking ahead, it is imperative to pursue further experimental validation through laboratory trials or field tests as a pivotal stride in fortifying the pragmatic utility of the models. Simulation-driven validation methodologies could, likewise, furnish invaluable insights into the real-world efficacy of the prognosticated properties.

Supplementary Information

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Additional file 1.

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Author contributions

Rodrigo Teixeira Schossler: conduction of data analyses, conceptualization, methodology, investigation, visualization, validation, writing—original draft preparation. Shafi Ullah: data analyses support, writing—review and editing. Zaid Alajlan: data analyses support, writing—review and editing. Xiong Yu: secure funding, conceptualization, design of overall analyses goal, supervision, writing—review and editing.

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Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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